



RTL2GDSII WORKSHOP

FLOORPLAN CREATION

in

PHYSICAL DESIGN

using

INTEGRATED CIRCUIT COMPILER (ICC2)

Only for chip to startup (C2S) Training



Agenda

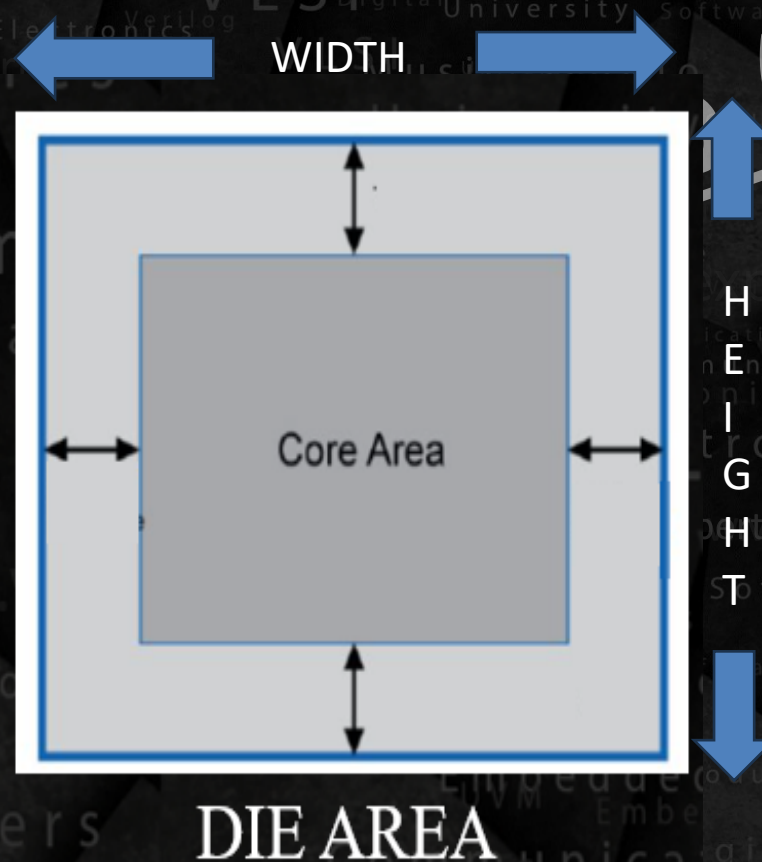
- Floorplan Basics
- Floorplan Inputs
- Design Planning Flow (Floorplan Creation Settings)
- Hands-On ICC2

Only for Chip to Startup (C2S) Training

Floorplan Basics

Floor planning – Stage 1 of Physical Design

- Define the **Shape** and **Size** of the block (Size depends on aspect ratio : W/H dimension)
- Floorplan also represents the **tentative placement** of all cells in design.
- It includes **Standard cell placement**, **Macro placement** and **IO Port placement** in design.



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(C2S) Training



Floorplan Basics – Components of Design

- **STD. CELLS** – FUNDAMENTAL COMPONENTS/LEAF CELLS
 - * PRE DEFINED, PRE TESTED & PRE CHARACTERIZED CELLS
 - * Eg. **NOT, AND, OR, Univ Gates, Complex Gates, Flip-Flop, Latches**
 - * Source : .lib/.db
- **MACROS** -- MEGA FUNCTIONAL BLOCKS
 - * HIGHLY OPTIMIZED (PPA)
 - * Eg. **MEMORY, PLL, ADC, DAC**
 - * Source : IP Lib

- **IO PADS** -- **Interface** with outside world (DATA I/O)
- **POWER PADS** – **SUPPLY** TO LOGICS in Core Area

Source : .lib/.db

SIZE IN FLOORPLANNING : ASPECT RATIO

Aspect Ratio – defines SIZE & SHAPE of a CHIP

$$\text{Aspect Ratio (Ar)} = \frac{\text{Width}}{\text{Height}}$$

Eg. If you specify a ratio of **1.00** (i.e., the height and width are the same) the core is a “**square**” shape.

- If you specify a ratio of **2.00**, Width is two times the Height, **Rectangular shape**.
- If **AR is 0.5**, Height is two times the width, **Rectangular shape**.

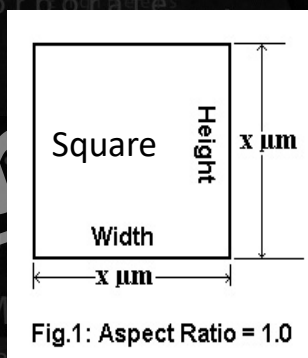
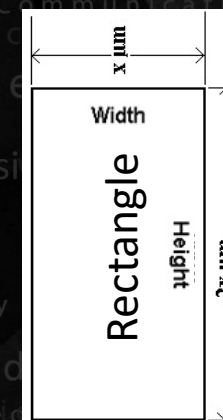
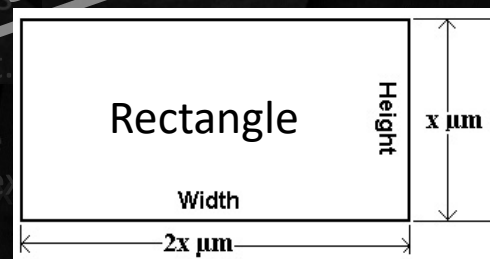


Fig.1: Aspect Ratio = 1.0



FLOORPLAN INPUTS

INPUT FILES

- LIB file (.ndm) – from Library Vendor/Synopsys
- MAPPED NETLIST file (.v) - Using Target Library file – From Design Compiler
- FP Creation file –script (.tcl)
- SDC file (.sdc) – (optional – Primary focus to place components only)

From Design Engineer



DESIGN PLANNING FLOW

FLOORPLAN SETTINGS

1. Create_design_library

2. Read_Verilog_netlist

3. Initialize_floorplan

4. Constraints to Place I/Os

5. Create_Placement

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